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Isothermal Elastohydrodynamic Lubrication of Point Contacts

Part III—Fully Flooded Results

Utilizing the theory developed by the authors in an earlier publication, the influence of the ellipticity parameter, the dimensionless speed, load, and material parameters on minimum film thickness was investigated. The ellipticity parameter was varied from one (a ball on a plate configuration) to eight (a configuration approaching a line contact). The dimensionless speed parameter was varied over a range of nearly two orders of magnitude. The dimensionless load parameter was varied over a range of one order of magnitude. Conditions corresponding to the use of solid materials of bronze, steel, and silicon nitride and lubricants of paraffinic and naphthenic mineral oils were considered in obtaining the exponent in the dimensionless material parameter. Thirty-four different cases were used in obtaining the minimum film thickness formula given below as

$$\bar{H}_{min} = 3.63U^{0.68}G^{0.49}W^{-0.073}(1 - e^{-0.68k})$$

A simplified expression for the ellipticity parameter was found where

$$k = 1.03 \left(\frac{R_y}{R_x} \right)^{0.64}$$

Contour plots were also shown which indicate in detail the pressure spike and two side lobes in which the minimum film thickness occurs. These theoretical solutions of film thickness have all the essential features of the previously reported experimental observations based upon optical interferometry.

Introduction

Only in recent years has the complete theoretical solution of the isothermal elastohydrodynamic lubrication (EHL) of point contacts been successfully analyzed. The analysis requires the simultaneous solution of the elasticity and Reynolds equation. The authors' approach to the theoretical solution has been presented in two previous publications (references [1] and [2]).¹ The first of these references (reference [1]) presents the elasticity model in which the conjunction is divided into equal rectangular areas with a uniform pressure applied over each area. The second of these references (reference [2]) gives the complete approach to solving the elastohydrodynamic lubrication problem for point contacts.

¹ Numbers in brackets designate References at end of paper.

Contributed by the Lubrication Division and presented at the Joint Lubrication Conference, Boston, Mass., October 5-7, 1976, of THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS. Manuscript received at ASME Headquarters July 6, 1976. Paper No. 76-Lub-30.

The most important practical aspect of the EHL point contact theory (reference [2]) is the determination of the minimum film thickness within the contact. That is, maintaining a fluid film thickness of adequate magnitude is extremely important to the operation of some machine elements. In the present paper only the results from the theory given in references [1] and [2] will be presented. In the results the influence of contact geometry as expressed in the ellipticity parameter, dimensionless speed, load, and material parameters on minimum film thickness is investigated for a conjunction fully immersed in lubricant (i.e., fully flooded). The ellipticity results have been presented in reference [3] by the authors but in the present paper additional solutions are presented and the accuracy of the presentation is improved. The corrected ellipticity results will be presented in this paper. In the numerical work the ellipticity parameter was varied from one (a ball on a plate configuration) to eight (a configuration approaching a line contact). The dimensionless speed and load parameters are varied over a range of about two and one orders of magnitude, respectively. Conditions equivalent to using solid materials of bronze, steel, and silicon nitride, and lubricants of paraffinic and naphthenic mineral oils were considered in obtaining the exponent on the dimensionless material parameter. Thirty-four different

cases were used in obtaining the fully flooded film thickness formula. A fully flooded condition is said to exist when the inlet distance of the conjunction ceases to influence in any significant way the minimum film thickness. The inlet distance of the conjunction is defined as the distance from the center of the contact to the edge of the computing area. Contour plots are also shown that indicate in detail the pressure spike and two side lobes in which the minimum film thickness occurs.

Dimensionless Grouping

From the variables of the numerical analysis (reference [2]) the following dimensionless groups can be defined.

$$1 \quad \text{Dimensionless film thickness: } H = \frac{h}{R_x} \quad (1)$$

$$2 \quad \text{Ellipticity parameter: } k = \frac{a}{b} \quad (2)$$

$$3 \quad \text{Dimensionless speed parameter: } U = \frac{\eta_0 u}{E' R_x} \quad (3)$$

$$4 \quad \text{Dimensionless load parameter: } W = \frac{F}{E' R_x^2} \quad (4)$$

$$5 \quad \text{Dimensionless material parameter: } G = \frac{E'}{p_{iv,as}} \quad (5)$$

Table 1 Effect of ellipticity parameter on minimum film thickness

Ellipticity parameter, k	Minimum film thickness		Difference between $H_{\min}$ and $\tilde{H}_{\min}$, D_1 , percent
	Obtained from EHL point-contact theory, $H_{\min}$	Obtained from least-square fit, $\tilde{H}_{\min}$	
1	3.367×10^{-6}	3.464×10^{-6}	+2.88
1.25	4.105	4.031	-1.80
1.5	4.565	4.509	-1.22
1.75	4.907	4.913	+0.11
2	5.255	5.252	-0.05
2.5	5.755	5.781	+0.45
3	6.091	6.156	+1.08
4	6.636	6.613	-0.34
6	6.969	6.961	-0.12
8	7.048	7.050	+0.02

Nomenclature

a = semimajor axis of contact ellipse

b = semiminor axis of contact ellipse

$$D_1 = \left(\frac{\tilde{H}_{\min} - H_{\min}}{H_{\min}} \right) 100$$

$$D_2 = \left(\frac{\tilde{H}_c - H_c}{H_c} \right) 100$$

E = modulus of elasticity

$$E' = \frac{2}{\left(\frac{1 - \nu_A^2}{E_A} + \frac{1 - \nu_B^2}{E_B} \right)}$$

F = normal applied load

G = dimensionless material parameter, $E'/p_{iv,as}$

h = film thickness

H = dimensionless film thickness, h/R_x

H_c = dimensionless central film thickness obtained from the EHL point contact theory (reference [2])

$\tilde{H}_c$ = dimensionless central film thickness obtained from the least square fit of data

$H_{\min}$ = dimensionless minimum film thickness obtained from EHL point contact theory (reference [2])

$\tilde{H}_{\min}$ = dimensionless minimum film thickness obtained from least square fit of data

$H_{\min,L}$ = dimensionless film thickness for line contact

k = ellipticity parameter, a/b

P = dimensionless pressure, p/E'

p = pressure

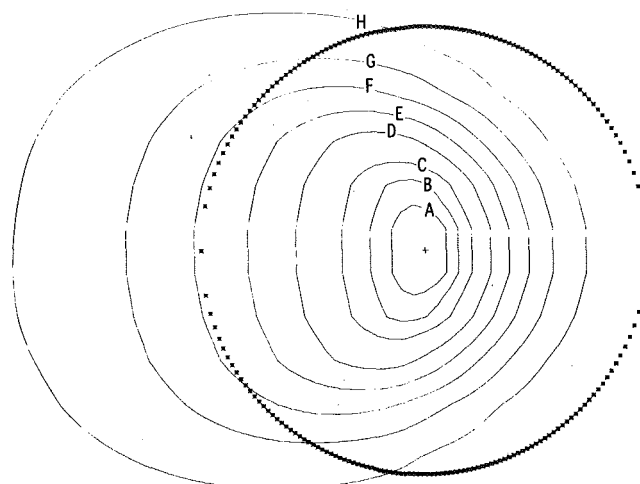
$p_{iv,as}$ = asymptotic isoviscous pressure

R = effective radius

r = radius of curvature

Dimensionless pressure, $P = p/E'$

A 0.70×10^{-3}
B .66
C .60
D .50
E .40
F .30
G .20
H .10



(a) Contour plot of dimensionless pressure.

Fig. 1 Contour plots of dimensionless pressure for ellipticity parameter (k) of 8 and 1.25, respectively. The dimensionless parameters U , W , and G are held constant as defined in equation (7).

where $p_{iv,as}$ = asymptotic isoviscous pressure obtained from Roelands (reference [4])

The asymptotic isoviscous pressure can be approximated by the inverse of the pressure viscosity coefficient ($p_{iv,as} \approx 1/\alpha$).

The dimensionless film thickness can be written as

$$H = f(k, U, W, G) \quad (6)$$

The most important practical aspect of elastohydrodynamic lubrication point contact theory developed in reference [2] is the determination of the minimum film thickness within the conjunction.

U = dimensionless speed parameter, $\frac{u \eta_0}{E' R_x}$

u = surface velocity in X direction

W = dimensionless load parameter, $\frac{F}{E' R_x^2}$

$x, X, \bar{X}$
 $y, Y, \bar{Y}$ = coordinate systems defined in paper

α = pressure viscosity coefficient

η_0 = atmospheric viscosity

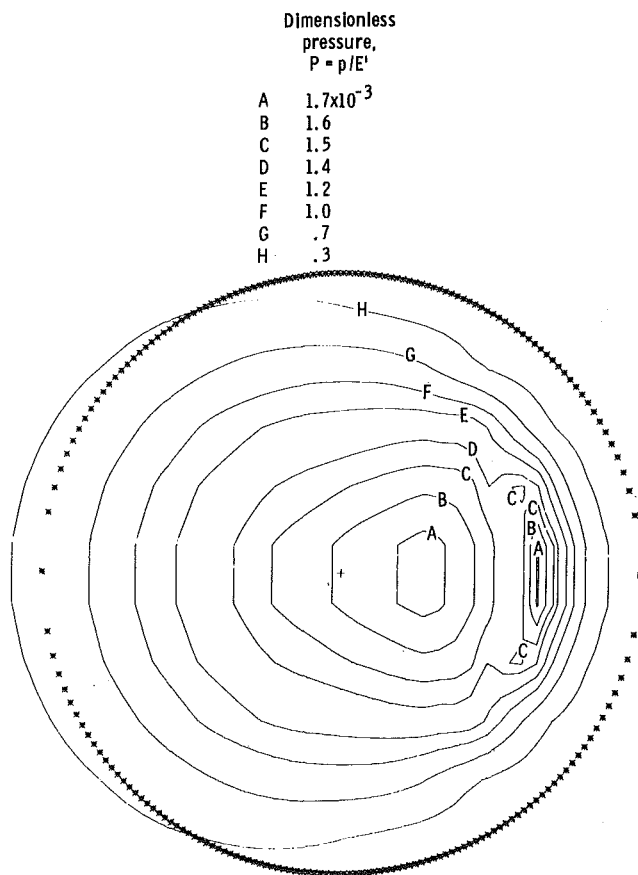
ν = Poisson ratio

Subscripts

A = solid A

B = solid B

x, y = coordinates system defined in paper



(b) Contour plot of dimensionless pressure.

Fig. 1 (concluded)

Therefore, in the fully flooded results to be presented the dimensionless parameters (k , U , W , and G) will be varied and the effect upon the minimum film thickness will be studied.

The Effect of Ellipticity of the Elastic Conjunction

The ellipticity parameter (k) is a function of the radii of curvature of the solids only (r_{Ax} , r_{Bx} , r_{Ay} , and r_{By}). The radii of curvature in the x direction for both solids A and B are used in defining the dimensionless speed and load parameters. Therefore, only the radius of curvature of solid B in the y direction will be changed in varying the ellipticity parameter from one (a ball on a plate configuration) to eight (a configuration approaching a line contact). In doing this the dimensionless speed (U), load (W), and material (G) parameters were held constant at the following values

$$\begin{aligned} U &= 0.1683 \times 10^{-11} \\ W &= 0.1106 \times 10^{-6} \\ G &= 4522 \end{aligned} \quad (7)$$

Care was taken to ensure that the highest ellipticity parameter ($k = 8$) case was in the elastic region as defined by Dowson-Higginson (page 101, reference [5]). For values of ellipticity parameter less than eight the results move farther into the elastic region.

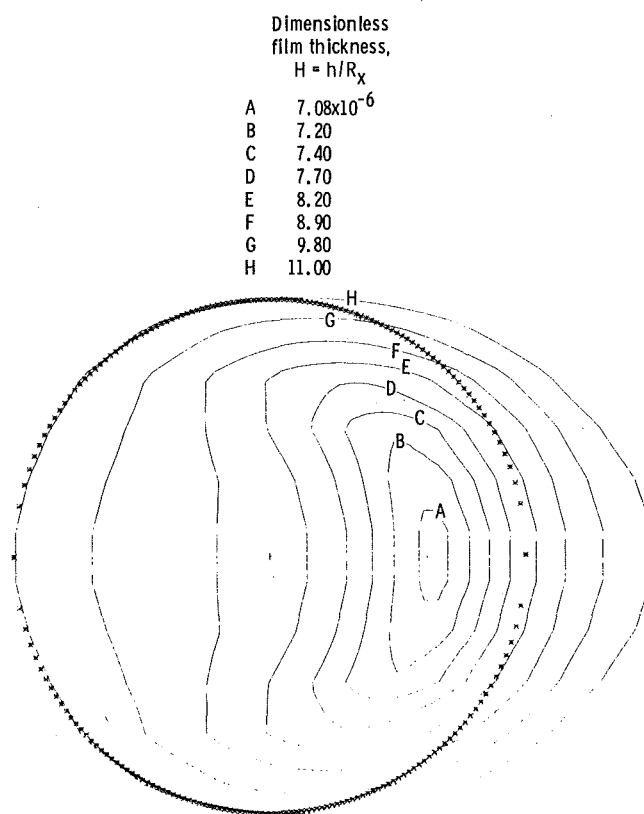
Table 1 gives ten values of ellipticity parameter (k) and the corresponding minimum film thickness ($H_{\min}$) as obtained from the EHL point contact theory (reference [2]). Having these ten pairs of data the object was to determine an equation that describes how the ellipticity parameter affects the minimum film thickness. The general form of this equation can be written as

Table 2 Effect of dimensionless speed parameter on minimum film thickness

Dimensionless speed parameter, U	Minimum film thickness		Difference between $H_{\min}$ and $\tilde{H}_{\min}$, percent
	Obtained from EHL point-contact theory, $H_{\min}$	Obtained from least-square fit, $\tilde{H}_{\min}$	
0.08416×10^{-11}	3.926×10^{-6}	3.915×10^{-6}	-0.275
.1683	6.156	6.252	+1.564
.2525	8.372	8.223	-1.780
.3367	9.995	9.987	-.078
.4208	11.61	11.61	-.004
.5892	14.39	14.57	+1.280
.8416	18.34	18.54	+1.104
1.263	24.47	24.39	-.320
1.683	29.75	29.61	-.467
2.104	34.58	34.43	-.432
2.525	39.73	38.95	-1.977
2.946	43.47	43.22	-.576
3.367	47.32	47.30	-.042
4.208	54.57	54.99	+.765
5.050	61.32	62.20	+1.430

$$\left(1 - \frac{H_{\min}}{H_{\min,L}}\right) = Ae^{Bk} \quad (8)$$

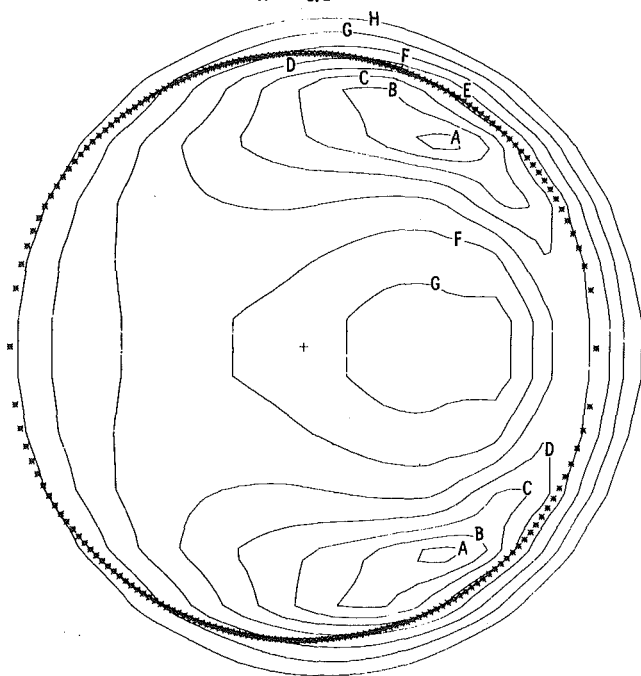
A least square exponential curve fit to the ten pairs of data points $\{[k_i, (1 - H_{\min}/H_{\min,L})_i], i = 1 \dots 10\}$ was used in obtaining values for A and B in equation (8). Besides a least square fit a coefficient of



(a) Contour plot of dimensionless film thickness.

Fig. 2 Contour plots of dimensionless film thickness for ellipticity parameter (k) of 8 and 1.25, respectively. The dimensionless parameters U , W , and G are held constant as defined in equation (7).

Dimensionless film thickness, $H = h/R_x$	
A	4.3×10^{-6}
B	4.6
C	5.0
D	5.5
E	6.0
F	6.6
G	7.4
H	8.2



(b) Contour plot of dimensionless film thickness.

Fig. 2 (concluded)

determination (r^2) is obtained. The value of r^2 reflects the fit of the data to the resulting equation: one being a perfect fit, and zero the worst possible fit. The minimum film thickness for a line contact ($H_{\min,L}$) used in equation (8) was determined by finding the $H_{\min,L}$ that gives a coefficient of determination closest to one. This value of $H_{\min,L}$ turned out to be 7.082×10^{-6} with a corresponding coefficient of determination of 0.9990, which is an excellent fit. Furthermore, the values of A and B in equation (8) as obtained from the least squared fit are

$$A = 0.9966 \approx 1.00 \quad (9)$$

$$B = -0.6752 \approx -0.68 \quad (10)$$

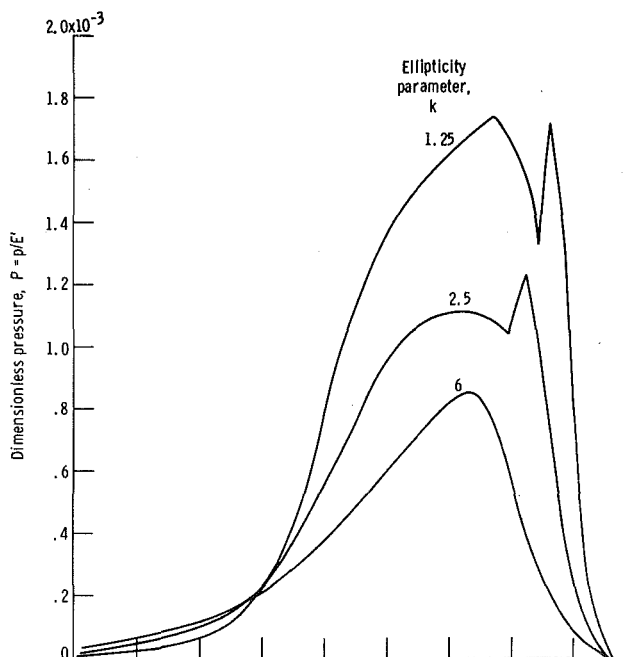
From equations (8), (9), and (10) the following equation can be written which shows the effect of ellipticity parameter on minimum film thickness.

$$\bar{H}_{\min} \propto (1 - e^{-0.68k}) \quad (11)$$

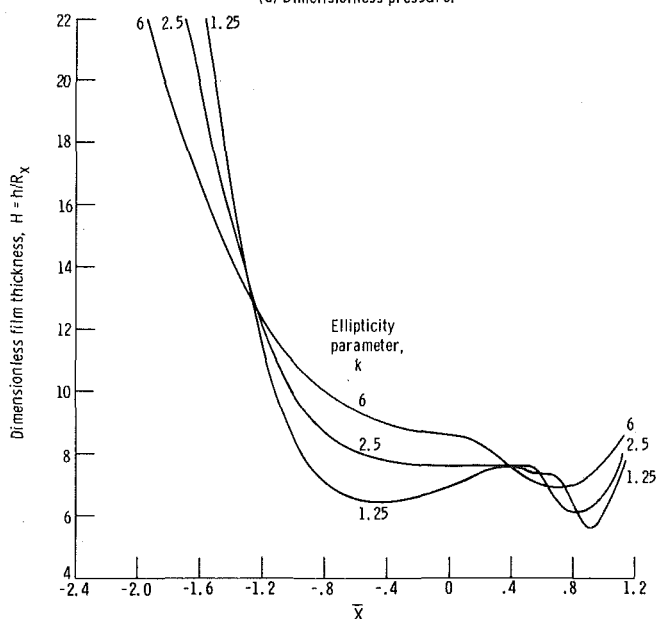
It is most significant that A turns out to be 0.9966 or approximately 1.00 since as $k \rightarrow 0$, $H_{\min} \rightarrow 0$. Therefore, even though the smallest value of k used in obtaining equation (11) was unity, it would seem that equation (11) could be applied to smaller values since in the limiting case ($k \rightarrow 0$) equation (11) satisfies the physical intuition. For the other extreme of large (k), a line contact situation is approached and the agreement with existing results is again good. From Dowson and Higginson (reference [5]) the line contact minimum thickness for the dimensionless parameters given in equation (7) is 7.720×10^{-6} . Compare this with 7.082×10^{-6} from the present results. The dif-

ference of 9 percent could well be the result of Dowson-Higginson (reference [5]) using an exponential pressure-viscosity relation instead of the Roelands (reference [4]) formulation used in the present work. As was pointed out in the closure of reference [2] (answering a query by P. M. Ku) the Roelands formula (reference [4]) suppresses the pressure spike somewhat and also results in a smaller film thickness.

Substituting equations (9) and (10) into equation (8) gives $\bar{H}_{\min}$, the dimensionless minimum film thickness obtained from the least square formulation. The $\bar{H}_{\min}$ for the ten values of ellipticity parameters are given in Table 1. The percentage difference between the minimum film thickness obtained from EHL point contact theory



(a) Dimensionless pressure.



(b) Dimensionless film thickness.

Fig. 3 Variation of dimensionless pressure and film thickness on $\bar{X}$ -axis for three values of ellipticity parameter. The value of Y is held fixed near axial center of contact.

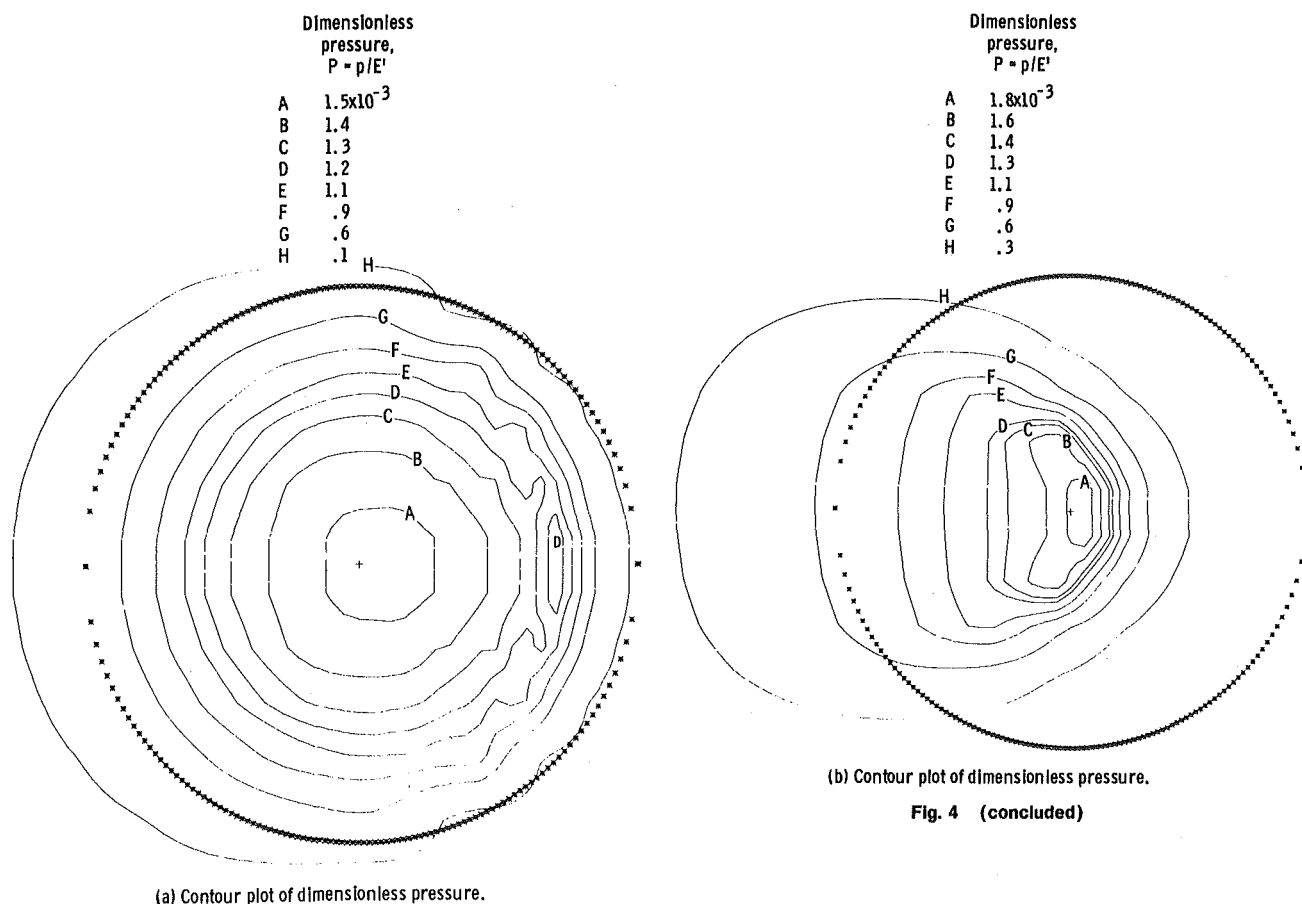


Fig. 4 Contour plots of dimensionless pressure for dimensionless speed parameter (U) of 8.416×10^{-13} and 0.5050×10^{-10} , respectively. The dimensionless parameters k , W , and G are held constant as defined in equation (13).

($H_{\min}$) and the minimum film thickness from the least squared fit equation ($\tilde{H}_{\min}$) is expressed as

$$D_1 = \left(\frac{\tilde{H}_{\min} - H_{\min}}{H_{\min}} \right) 100 \quad (12)$$

Note that in Table 1, the magnitude of D_1 is within ± 3 percent for all cases.

Figs. 1(a) and (b) give contour plots of dimensionless pressure for two extreme values of the ellipticity parameters ($k = 8$ and $k = 1.25$). In these and all contour plots to be presented the + symbol indicates the center of the Hertzian contact. Note that, because of the dimensionless representation of the X and Y coordinates, the actual Hertzian contact ellipse becomes a circle regardless of the value of the ellipticity parameter. The Hertzian contact circle is shown in each figure by asterisks. At the top of each figure the contour label and the corresponding value of dimensionless pressure are given. The inlet region is to the left and the exit region is to the right.

For an ellipticity parameter of eight the maximum pressure is seen to be near the center of the contact, and even though the conditions are in the elastic region no pressure spike is seen to occur. The pressure gradient at the exit end of the conjunction is much larger than at the inlet region. For a low value of the ellipticity parameter of 1.25 a pressure spike is visible at the exit end of the contact.

Figs. 2(a) and (b) show two contour plots of film thickness when the ellipticity parameter (k) is 8 and 1.25, respectively. For the ellipticity parameter of eight the minimum film thickness is directly behind the axial center of the contact. For the ellipticity parameter of 1.25 two minimum film thickness areas occur to the sides and nearer the edge of the Hertzian circle. These results showing the two "side lobes" in which minimum film thickness areas occur produce all the

essential features of previously reported experimental observations based upon optical interferometry (reference [6]).

Figs. 3(a) and (b) show the variation of pressure and film thickness respectively in the $\bar{X}$ direction close to the mid-plane of the contact for three values of the ellipticity parameter. As has been true for all the ellipticity parameter results presented, the value of the dimensionless speed, load, and material parameters were held fixed as per equation (7). In Fig. 3(a) we find that for $k = 6$ no spike occurs, but this well-known feature of theoretical solution to the EHL problem is evident for $k = 2.5$ and $k = 1.25$.

In Fig. 3(b) for $k = 1.25$ the central region is not parallel with the $\bar{X}$ axis. The reason is probably that compressibility effects are considered in the theory (reference [2]). That is, when compressibility is considered the film thickness in the center is reduced by the amount that the fluid volume decreases at high pressure (reference [5]).

Table 3 Effect of dimensionless load parameter on minimum film thickness

Dimensionless load parameter, W	Minimum film thickness		Difference between $H_{\min}$ and $\tilde{H}_{\min}$, D_1 , percent
	Obtained from EHL point-contact theory, $H_{\min}$	Obtained from least-square fit, $\tilde{H}_{\min}$	
0.1106×10^{-6}	6.969×10^{-6}	6.941×10^{-6}	-0.41
.2211	6.492	6.599	+1.65
.3686	6.317	6.358	+.64
.5528	6.268	6.172	-1.52
.7371	6.156	6.044	-1.81
.9214	6.085	5.947	-2.27
1.106	5.811	5.868	+.98
1.290	5.657	5.803	+2.58

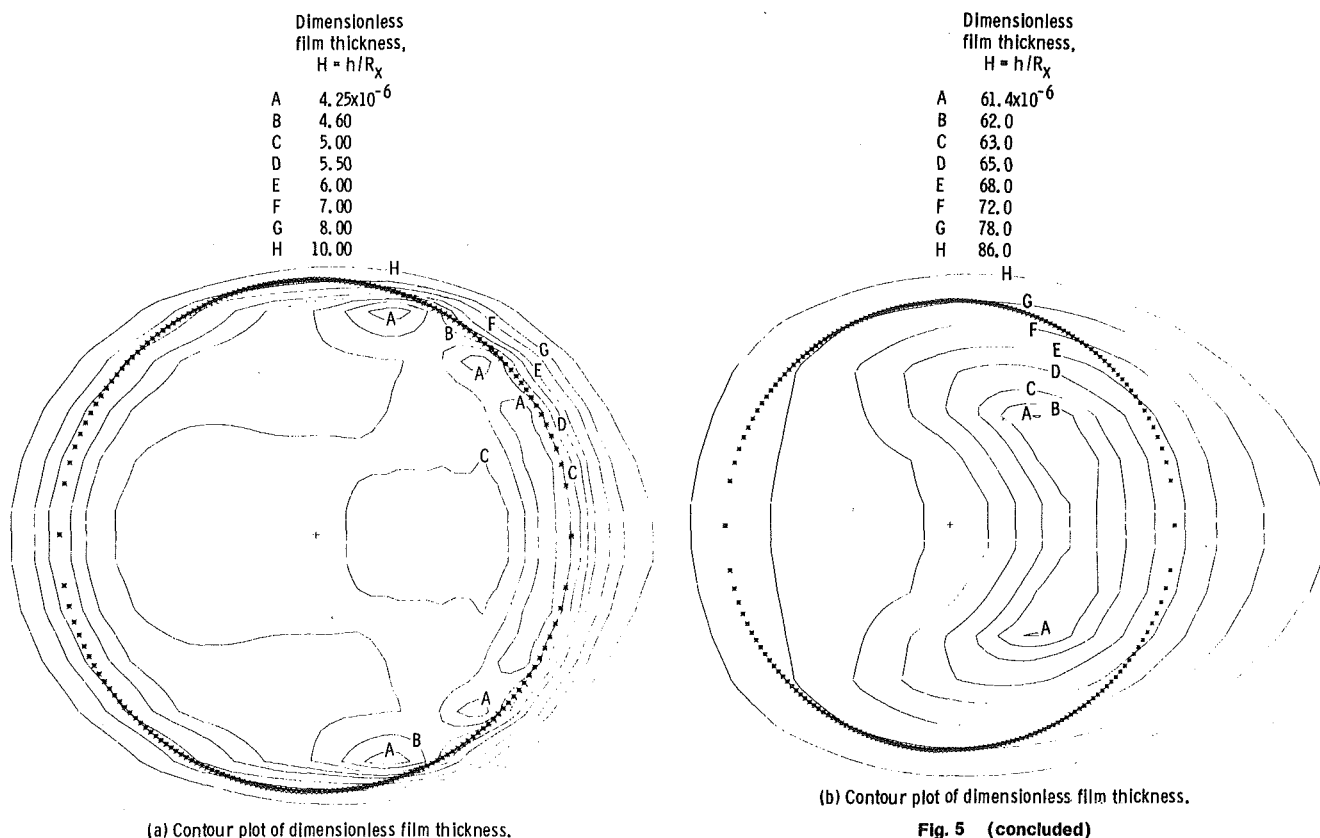


Fig. 5 Contour plots of dimensionless film thickness for dimensionless speed parameter (U) of 0.8416×10^{-12} and 5.05×10^{-11} . The dimensionless parameters k , W , and G are held constant as defined in equation (13).

The Influence of Speed

By changing only the surface velocity in the x direction (u) the dimensionless speed parameter (U) (equation (3)) changes, but the other dimensionless parameters (k , W , and G) remain constant. The values at which these dimensionless parameters were held constant in the calculations are

$$\left. \begin{aligned} W &= 0.7371 \times 10^{-6} \\ G &= 4522 \\ k &= 6 \end{aligned} \right\} \quad (13)$$

Table 2 gives the dimensionless speed parameter (U) and the corresponding minimum film thickness ($H_{\min}$) as obtained from the EHL point contact theory (reference [2]). There are fifteen different values of the dimensionless speed parameter covering nearly two orders of magnitude. Having these fifteen pairs of data the objective is to determine an equation that describes how the dimensionless speed affects the minimum film thickness. The general form of this equation can be written as

$$H_{\min} = IU^J \quad (14)$$

By applying a least square power fit to the fifteen pairs of data $\{(U_i, H_{\min,i}) \mid i = 1 \dots 15\}$ the values of I and J were found to be

$$I = 560.18 \quad (15)$$

$$J = 0.67542 \approx 0.68 \quad (16)$$

The coefficient of determination (r^2) for these results was excellent at 0.9998. Substituting equations (15) and (16) into equation (14) gives the values of $\tilde{H}$ shown in Table 2. The percentage difference (D_1) between the minimum film thickness obtained from the EHL point contact theory ($H_{\min}$) and the minimum film thickness obtained from the least squared fit ($\tilde{H}_{\min}$) is expressed in equation (12) and given

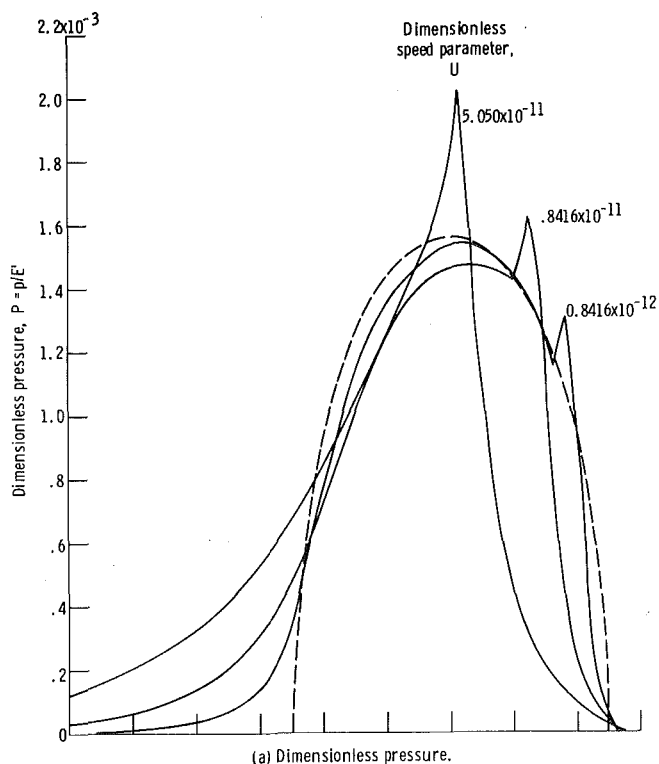
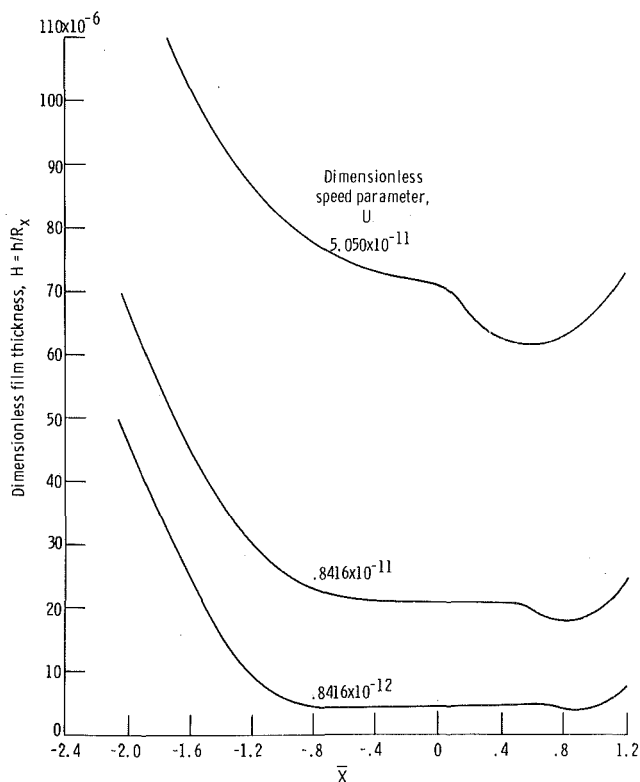


Fig. 6 (a) Variation of dimensionless pressure and film thickness on $\bar{X}$ -axis for three values of dimensionless speed parameter. The value of Y is held fixed near axial center of contact.



(b) Dimensionless film thickness.

Fig. 6 (b) Dimensionless film thickness

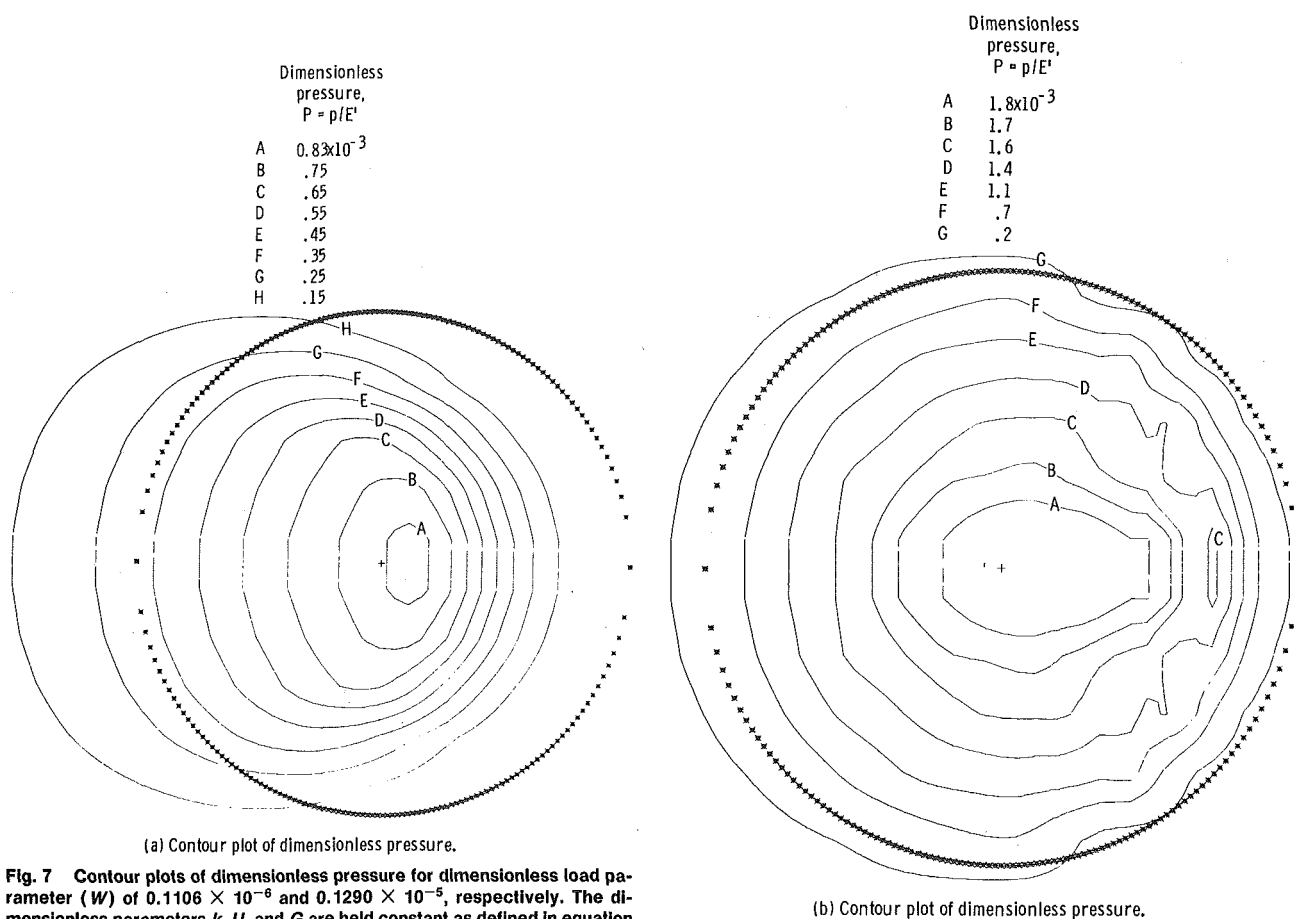
in Table 2. Note that the variation of D_1 is less than ± 2 percent.

From equations (14) and (16) the effect of dimensionless speed on dimensionless minimum film thickness can be written as

$$\bar{H}_{\min} \propto U^{0.68} \quad (17)$$

Figs. 4(a) and (b) give contour plots of dimensionless pressure for two extreme values of dimensionless speed parameters ($U = 0.8416 \times 10^{-12}$ and $U = 0.5050 \times 10^{-10}$). For the low speed case ($U = 0.8416 \times 10^{-12}$) we see a pressure spike at the exit end of the contact. In the high speed case ($U = 0.5050 \times 10^{-10}$) no pressure spike occurs. It should be noted from Figs. 4(a) and (b) that the pressure in the inlet region is higher at high speeds than at low speeds.

Figs. 5(a) and (b) show contour plots of dimensionless film thickness when the dimensionless speed is equal to 0.8416×10^{-12} and 0.5050×10^{-10} , respectively. In Figs. 5(a) the minimum film thickness appears close to the Hertzian circle and off to the side. In Fig. 5(b) the minimum film thickness area appears between the center of the contact and the Hertzian circle. Figs. 6(a) and (b) show the variation of pressure and film thickness, respectively, on the $\bar{X}$ axis near the mid-plane of the conjunction for three values of dimensionless speed parameter. As has been true for all the speed results the values of the dimensionless load, material, and ellipticity parameter are held fixed as described in equation (13). In Fig. 6(a) the dashed line corresponds to the Hertzian pressure distribution. It can be seen from Fig. 6(a) that the pressure in the inlet region is higher for the high speed ($U = 0.5050 \times 10^{-10}$) profile. For $U = 0.8416 \times 10^{-12}$ the pressure spike is seen to originate very near the trailing edge to the Hertzian pressure and as the speed increases the pressure spike moves upstream. For line contacts Dowson and Higginson (reference [5]) found similar results to those shown in Fig. 6(a).



(a) Contour plot of dimensionless pressure.

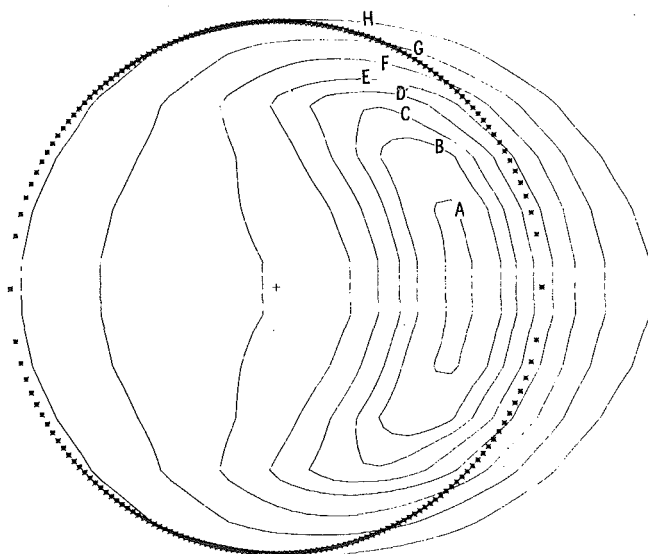
(b) Contour plot of dimensionless pressure.

Fig. 7 (concluded)

Fig. 7 Contour plots of dimensionless pressure for dimensionless load parameter (W) of 0.1106×10^{-6} and 0.1290×10^{-5} , respectively. The dimensionless parameters k , U , and G are held constant as defined in equation (18).

Dimensionless
film thickness,
 $H = h/R_x$

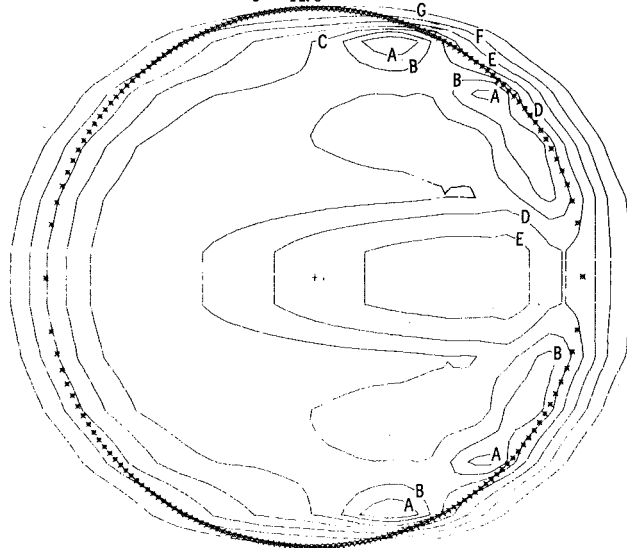
A 7.0×10^{-6}
B 7.2
C 7.4
D 7.7
E 8.1
F 8.7
G 9.5
H 10.7



(a) Contour plot of dimensionless film thickness.

Dimensionless
film thickness,
 $H = h/R_x$

A 6.1×10^{-6}
B 6.5
C 7.0
D 7.7
E 8.7
F 10.0
G 12.0



(b) Contour plot of dimensionless film thickness.

Fig. 8 (concluded)

Fig. 8 Contour plots of dimensionless film thickness for dimensionless load parameter (W) of 0.1106×10^{-6} and 0.1290×10^{-5} , respectively. The dimensionless parameters k , U , and G are held constant as defined in equation (18).

The typical elastohydrodynamic film shape with an essentially parallel section in the central region is shown in Fig. 6(b). Also there is a considerable change in film thickness as the dimensionless speed is changed, as indicated by equation (17). This illustrates most clearly the dominant effect of the dimensionless speed parameter (U) upon the minimum film thickness in elastohydrodynamic contacts.

The Influence of Load

By changing only the normal applied load (F) in equation (4) the dimensionless load parameter (W) changes while the remaining dimensionless parameters (k , U , and G) remain constant. The values at which these parameters were held constant are given below.

$$\left. \begin{aligned} U &= 0.1683 \times 10^{-11} \\ G &= 4522 \\ k &= 6 \end{aligned} \right\} \quad (18)$$

Table 3 gives the dimensionless load parameter (W) and the corresponding minimum film thickness ($H_{\min}$) as obtained from the EHL point contact theory (reference [2]). There are eight different values of the dimensionless load parameters covering over an order of magnitude. Having these eight pairs of data the objective was to determine an equation which describes how the dimensionless load affects the minimum film thickness. The general form of this equation can be written as

$$H_{\min} = KW^L \quad (19)$$

By applying a least square power fit to the eight pairs of data (W_i , $H_{\min,i}$), $i = 1 \dots 8$ the values of K and L were found to be

$$K = 2.1592 \times 10^{-6} \quad (20)$$

$$L = -0.072924 \approx -0.073 \quad (21)$$

The coefficient of determination (r^2) for these results was 0.9260, which was good, but was the lowest obtained in deriving the minimum film thickness equation (equation (28)). Substituting equations (20) and (21) into equation (19) gives the values of $\bar{H}_{\min}$ shown in Table

Table 4 Effect of solid material and lubricant as represented in dimensionless material parameter on minimum film thickness

Solid material	Lubricant	Dimensionless material parameter, G	Dimensionless speed parameter, U	Dimensionless load parameter, W	Minimum film thickness		Difference between $H_{\min}$ and $\bar{H}_{\min}$, percent
					Obtained from EHL point-contact theory, $H_{\min}$	Obtained from least-square fit, $\bar{H}_{\min}$	
Bronze	Paraffinic	2310	0.3296×10^{-11}	0.7216×10^{-6}	6.931×10^{-6}	6.873×10^{-6}	-0.84
Bronze	Naphthenic	3591	.9422	.7216	17.19	17.404	+1.25
Steel	Paraffinic	4522	.1683	.3686	6.317	6.336	+0.31
Silicon nitride	Paraffinic	6785	.1122	.2456	6.080	6.038	-0.70

3. The percentage difference (D_1) between the minimum film thickness obtained from the EHL point contact theory ($H_{\min}$) and the minimum film thickness from the least squared fit equation ($\bar{H}_{\min}$) is expressed in equation (12) and given in Table 3. In Table 3 note that variation of D_1 is within ± 3 percent at all cases.

From equations (19) and (21) the effect of load on minimum film thickness can be written as

$$\bar{H}_{\min} \propto W^{-0.073} \quad (22)$$

Figs. 7(a) and (b) give contour plots of dimensionless pressure for the two extreme values of dimensionless load parameter ($W = 0.1106 \times 10^{-6}$ and $W = 0.1290 \times 10^{-5}$) that were investigated. No pressure spike is evident at the lowest load, whereas at the highest load a pressure spike is visible.

Contour plots of dimensionless film thickness for the same two values of dimensionless load parameters as given in Fig. 7 are shown in Fig. 8. In Fig. 8(a) for the low load case ($W = 0.1106 \times 10^{-6}$) the minimum film thickness occurs directly behind the center of the contact. In Fig. 8(b) for the high load ($W = 0.129 \times 10^{-5}$) the minimum film thickness is off to the sides in two areas close to the Hertzian circle.

The variation of pressure and film thickness in the $\bar{X}$ direction along a line close to the mid-plane of the conjunction is shown in Fig. 9 for three values of the dimensionless load parameter. The values of the dimensionless speed, material, and ellipticity parameters were held fixed as described by equation (18) for all computations at various loads. In Fig. 9(a) note that as the dimensionless load is increased the inlet pressure becomes smaller. For the highest load case shown in Fig. 9(b) film thickness rises between the central region and the outlet restriction in the same manner as seen in Fig. 3(b). Again this is attributed to compressibility effects of the fluid. Also note that at a load of $W = 0.5528 \times 10^{-6}$ the film thickness is slightly smaller than at a load of $W = 1.06 \times 10^{-6}$. The reason is that at the lower load the minimum film thickness is closer to the axial center of the contact than at the higher load. As was pointed out in discussing Figs. 8(a) and (b), the location of the minimum film-thickness region changes as the dimensionless load is changed.

The Effect of Material Properties

Contrary to what was found in the previous three sections, the effect of the dimensionless material parameter on minimum film thickness is not a simple matter. As can be seen from equations (3), (4), and (5) when either the material of the solids (as expressed in E') or the lubricant (as expressed in η_0 and $p_{iv,as}$) is varied, not only does the material parameter (G) change, but so do the dimensionless speed (U) and load (W) parameters. Only the ellipticity parameter can be held fixed. For all the results presented in this section the ellipticity parameter is held fixed at a value of six.

Table 4 gives the four material parameter results. The general form showing how the minimum film thickness is a function of the dimensionless material parameter is given as

$$C = TG^V \quad (23)$$

where

$$C = \frac{H_{\min}}{(1 - e^{-0.68k}) U^{0.68} W^{-0.073}} \quad (24)$$

Note in equation (24) that the exponents are rounded off to two significant figures so that any error could be absorbed in T given in equation (23). By applying a least square power fit to the four pairs of data the values of T and V were found to be

$$T = 3.6891 \quad (25)$$

$$V = 0.48669 \approx 0.49 \quad (26)$$

The coefficient of determination for these results was 0.9980, which is excellent. Substituting equations (25) and (26) into equation (23) gives the values of $\bar{H}_{\min}$ shown in Table 4. The percentage difference (D_1) shown in Table 4 varies by less than 2 percent at all cases. Therefore, from equations (23) and (25) the effect of dimensionless

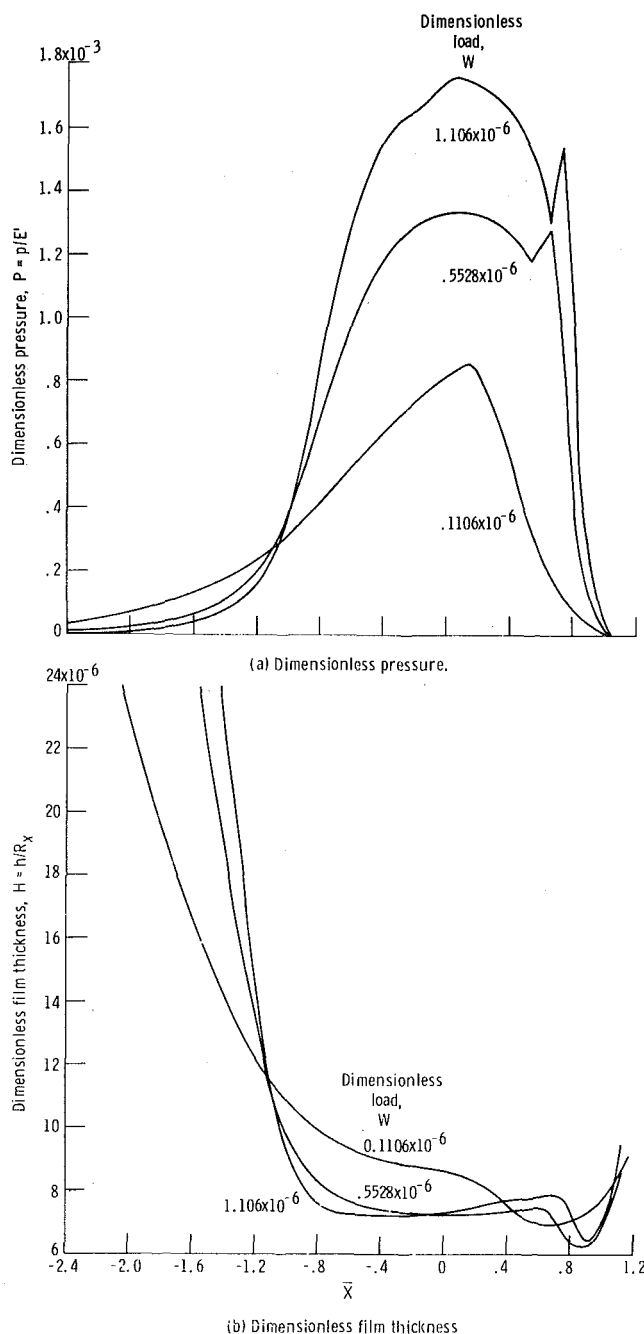


Fig. 9 Variation of dimensionless pressure and film thickness on $\bar{X}$ -axis for three values of dimensionless load parameter. The value of Y is held fixed near axial center of contact.

material parameter on dimensionless film thickness can be written as

$$\bar{H}_{\min} \propto G^{0.49} \quad (27)$$

Minimum Film Thickness Formula

The proportionality expressions (11), (17), (22), and (27) established how the minimum film thickness varies with the ellipticity, speed, load, and material parameters, respectively. This enables a composite minimum film thickness formula for fully flooded isothermal, elastohydrodynamic point contacts to be written as

$$\bar{H}_{\min} = 3.63 U^{0.68} G^{0.49} W^{-0.073} [1 - (e^{-0.68k})] \quad (28)$$

In equation (28) the constant 3.63 is different from that in equation

Table 5 Data showing effect of ellipticity, load, speed, and material on minimum film thickness

Case	Ellipticity parameter, k	Dimensionless load parameter, W	Dimensionless speed parameter, U	Dimensionless material parameter, G	Minimum film thickness		Difference between $H_{\min}$ and $\tilde{H}_{\min}'$, D_1 , percent	Results
					Obtained from EHL point-contact theory, $H_{\min}$	Obtained from least-square fit, $\tilde{H}_{\min}$		
1	1	0.1108×10^{-6}	0.1683×10^{-11}	4522	3.367×10^{-6}	3.514×10^{-6}	+4.37	Ellipticity
2	1.25				4.105	4.078	-.66	
3	1.5				4.565	4.554	-.24	
4	1.75				4.907	4.955	+ .98	
5	2				5.255	5.294	+ .74	
6	2.5				5.755	5.821	+1.15	
7	3				6.091	6.196	+1.72	
8	4				6.636	6.652	+ .24	
9	6				6.969	7.001	+ .46	
10	8				7.048	7.091	+ .61	
11	6	.2211			6.492	6.656	+2.53	Load plus case 9
12		.3686			6.317	6.412	+1.50	
13		.5528			6.268	6.225	-.69	
14		.7371			6.156	6.095	-.99	
15		.9214			6.085	5.997	-1.45	
16		1.106			5.811	5.918	+1.84	
17		1.290			5.657	5.851	+3.43	
18		.7371	.08416		3.926	3.805	-3.08	
19			.2525		8.372	8.032	-4.06	
20			.3367		9.995	9.769	-2.26	
21			.4208		11.61	11.37	-2.07	Speed plus case 14
22			.5892		14.39	14.29	-.69	
23			.8416		18.34	18.21	-.71	
24			1.263		24.47	24.00	-1.92	
25			1.683		29.75	29.18	-1.92	
26			2.104		34.58	33.96	-1.79	
27			2.525		39.73	38.44	-3.25	
28			2.946		43.47	42.69	-1.79	
29			3.367		47.32	46.76	-1.18	
30			4.208		54.57	54.41	-.29	
31			5.050		61.32	61.59	+ .44	Materials plus case 9
32		.7216	.3296	2310	6.931	6.938	+ .10	
33		.7216	.9422	3491	17.19	17.59	+2.33	
34		.2456	.1122	6785	6.080	6.116	+ .59	

(25) to account for rounding off the material parameter exponent.

Table 5 gives the thirty-four different cases used in obtaining equation (28). In this table $H_{\min}$ corresponds to the minimum film thickness obtained from the EHL point contact theory developed in reference [2] and $\tilde{H}_{\min}$ is the minimum film thickness obtained from equation (28). The percentage difference between these two values is expressed by D_1 , which is defined in equation (12). In Table 5 the values of D_1 are within ± 5 percent.

It is sometimes more convenient to express the side leakage factor in equation (28) in terms of the radius of curvature ratio (R_y/R_x) instead of the ellipticity parameter (k) through the following relationship.

$$k = 1.03 \left(\frac{R_y}{R_x} \right)^{0.64} \quad (29)$$

where

$$\frac{1}{R_y} = \frac{1}{r_{Ay}} + \frac{1}{r_{By}} \quad (30)$$

$$\frac{1}{R_x} = \frac{1}{r_{Ax}} + \frac{1}{r_{Bx}} \quad (31)$$

Using equation (29) avoids the need to evaluate elliptic integrals of the first and second kind in the determination of k . The minimum film thickness can thus be derived directly from a knowledge of the radii of curvature of the contacting bodies (r_{Ax} , r_{Bx} , r_{Ay} , and r_{By}).

It is interesting to compare the new point contact minimum film thickness formula (equation (28)) with the corresponding equation generated in the 1960s (reference [7]) for line contacts.

$$H_{\min,L} = 2.65 U^{0.70} G^{0.54} W^{-0.13} \quad (32)$$

The powers of U , G , and W in equations (28) and (32) are quite sim-

ilar, considering the different numerical procedures upon which they are based. It is also worth noting that the power of (W) in equation (28) is extremely close to the value of (-0.074) proposed by Archard and Cowling in their study of point contacts (reference [8]).

Central Film Thickness Formula

There is interest in knowing the central film thickness, in addition to the minimum film thickness in elastohydrodynamic contacts. The procedure used in obtaining the central film was the same as that used in obtaining the minimum film thickness and will not be repeated here. The central film thickness formula obtained from the results is given below as

$$\tilde{H}_c = 2.69 U^{0.67} G^{0.53} W^{-0.067} (1 - 0.61 e^{-0.73k}) \quad (33)$$

Comparing the central film thickness formula (equation (33)) with the minimum film thickness formula (equation (28)) reveals a slight difference. In equation (33) the load exponent is small but negative, as it was for the minimum film thickness formula. This is in contrast with the recent numerical study of Ranger, et al. (reference [9]) who found a small but positive exponent on the dimensionless load (W) in their formulation of a central film thickness.

Table 6 gives the thirty-four different cases used to obtain equation (33). In this table, H_c corresponds to the central film thickness obtained from the EHL point contact theory developed in reference [2], and $\tilde{H}_c$ corresponds to the central film thickness obtained from equation (33). The percentage difference between these two values is given by D_2 and is written as

$$D_2 = \left(\frac{\tilde{H}_c - H_c}{H_c} \right) 100 \quad (34)$$

In Table 6 the values of D_2 are within ± 10 percent.

Table 6 Data showing effect of ellipticity, load, speed, and material on central film thickness

Case	Ellipticity parameter, k	Dimensionless load parameter, W	Dimensionless speed parameter, U	Dimensionless material parameter, G	Central film thickness		Difference between H_c and H_c' , D_2 , percent	Results
					Obtained from EHL point-contact theory, H_c	Obtained from least-square fit, $\tilde{H}_c$		
1	1	0.1106×10^{-6}	0.1683×10^{-11}	4522	6.860×10^{-6}	6.215×10^{-6}	-9.40	Ellipticity
2	1.25				6.964	6.647	-4.55	
3	1.5				7.001	7.006	+0.07	
4	1.75				7.015	7.306	+4.15	
5	2				7.402	7.556	+2.08	
6	2.5				7.653	7.937	+3.71	
7	3				7.845	8.202	+4.55	
8	4				8.292	8.513	+2.67	
9	6				8.657	8.736	+0.91	Load plus case 9
10	8				8.672	8.787	+1.33	
11	6	.2211			7.796	8.339	+6.97	
12		.3686			7.505	8.059	+7.38	
13		.5528			7.309	7.843	+7.31	
14		.7371			7.517	7.693	+2.34	
15		.9214			7.611	7.578	-0.43	
16		1.106			7.416	7.487	+0.96	Speed plus case 14
17		1.290			6.762	7.410	+9.58	
18		.7371	.08416		4.917	4.836	-1.65	
19			.2525		9.999	10.10	+1.01	
20			.3367		11.40	12.24	+7.37	
21			.4208		13.07	14.21	+8.72	
22			.5892		17.13	17.81	+3.97	
23			.8416		21.35	22.61	+5.90	
24			1.263		29.62	29.68	+0.20	Materials plus case 9
25			1.683		35.50	35.98	+1.35	
26			2.104		41.05	41.79	+1.80	
27			2.525		46.64	47.22	+1.24	
28			2.946		51.08	52.36	+2.51	
29			3.367		55.56	57.26	+3.06	
30			4.208		63.81	66.49	+4.20	
31			5.050		71.25	75.13	+5.45	
32		.7216	.3296	2310	8.422	8.466	+0.52	Materials plus case 9
33		.7216	.9422	3591	20.70	21.62	+4.44	
34		.2456	.1122	6785	7.825	7.825	0	

Conclusion

By using the procedures outlined by the authors in an earlier publication (reference [2]) the influence of the ellipticity parameter and the dimensionless speed, load, and material parameters on minimum film thickness has been investigated. The ellipticity parameter was varied from one (a ball on a plate configuration) to eight (a configuration approaching a line contact). The dimensionless speed parameter was varied over a range of nearly two orders of magnitude. The dimensionless load parameter was varied over a range of one order of magnitude. Situations equivalent to using solid materials of bronze, steel, and silicon nitride, and lubricants of paraffinic and naphthenic mineral oils were considered in an investigation of the role of the dimensionless material parameter. Thirty-four different cases were used to generate the minimum film thickness and central film thickness relations.

$$\tilde{H}_{\min} = 3.63U^{0.68}G^{0.49}W^{-0.073}(1 - e^{-0.68k})$$

$$\tilde{H}_c = 2.69U^{0.67}G^{0.53}W^{-0.067}(1 - 0.61e^{-0.73k})$$

The ellipticity parameter (k) can be written as

$$k = 1.03 \left(\frac{R_y}{R_x} \right)^{0.64}$$

Contour plots have been presented that indicate in detail the pressure distribution and the film thickness. In some solutions, pressure spikes were in evidence. The theoretical solutions of film thickness have all the essential features of previously reported experimental observations based upon optical interferometry.

The importance of the present paper lies in the fact that it presents for the first time a complete theoretical film-thickness equation for elastohydrodynamic point contacts operating under fully flooded

conditions. The exponents on the various dimensionless parameters governing minimum film thickness in such conjunctions are quite similar to those developed earlier (see reference [7]) for line contacts. The most dominant exponent occurs in association with the speed parameter, while the exponent on the load parameter is very small and negative. The material parameter also carries a significant exponent, although the range of this parameter in engineering situations is limited. Ranger, et al. (reference [9]) have developed a central film thickness formula for the contact geometry of a ball on a plate from which an estimate can be made of the minimum film thickness. However, the present work is valid for any contact geometry and proceeds directly to the evaluation of the minimum film thickness.

Perhaps the most significant feature of the proposed minimum film thickness formula is that it can be applied to any contacting solids that present an elliptical Hertzian contact region. Many machine elements, particularly rolling element bearings, possess such geometry, and it is expected that the new minimum film thickness equation will find application in such fields.

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